

FlavorMap

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The Complete Project Bible

Computational Geography of Thai Cuisine · Network Analysis · Geographic Clustering · Interactive Visualization · Research Paper · 12-Month Execution

ISB Bangkok Grade 10 · Pillar 2 of the Future Leader Plan · University Admissions Data Science Portfolio

What this document is. This is the complete, authoritative bible for the FlavorMap project — everything needed to build it, research it, publish from it, and present it in a university application. It covers the research design, data collection strategy, full technical architecture, step-by-step implementation guide, interactive visualization specification, research paper plan, and 12-month timeline. Every section is written for a Grade 10 student who will implement this herself, guided by a coding mentor and an academic advisor.

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1. Why FlavorMap — The Research Case

"Food is the most intimate form of cultural transmission. Every dish carries the geography, history, and identity of the people who cook it. Thailand has 77 provinces, each with distinct culinary traditions. No machine has ever tried to map them. FlavorMap is that map."

The founding observation

Thai cuisine is described internationally as a single entity — 'Thai food.' But anyone who has eaten in Thailand knows this is wrong. A bowl of noodles in Chiang Mai tastes nothing like a bowl of noodles in Hat Yai. Nam prik in Isaan uses fermented fish paste (pla ra) that Central Thai cooks consider too pungent. Southern curries use fresh turmeric where Northern curries use dried ginger. These differences are not random. They encode centuries of trade routes, migration patterns, neighboring country influences, and ecological constraints. Nobody has formally mapped them at the ingredient level using data science.

The gap in existing research

- **Academic food science** studies Thai cuisine nutritionally or chemically — not geographically or culturally
- **Food media** (Wongnai, food blogs) describes regional dishes anecdotally — no quantitative comparison
- **Tourism materials** list 'regional specialties' without analyzing what makes them structurally distinct
- **Computational food science** (the Ahn et al. 2011 flavor network paper, West et al. 2013 regional cuisine paper) has studied Western and East Asian cuisines extensively — Thai cuisine has never been analyzed this way
- **The specific gap:** No province-level ingredient network analysis of Thai cuisine exists in any published form

Why this is the right second project for her

- **Genuine love of cooking.** She loves food and cooks Thai food. Her personal knowledge of ingredients is itself domain expertise that makes her a credible researcher in this space. She can judge whether the model's findings are meaningful.
- **Research-first, not product-first.** TunDee builds a product. FlavorMap asks a question and finds an answer. Having both shows she can operate in both modes — builder and researcher.
- **Different technical skills.** Every technical method in FlavorMap is new — graph theory, clustering, geospatial analysis, interactive visualization. Nothing repeats TunDee.
- **Publishable in respected venues.** Digital humanities, cultural analytics, and food computing are active research areas with real conferences. A novel finding about Thai culinary geography is publishable.
- **Produces a lasting public asset.** The dataset she builds has permanent value to researchers, food companies, and cultural organizations. It is not a product that needs maintenance — it is data infrastructure.

The one-sentence pitch to admissions readers

"I used network analysis and geographic clustering to build the first computational map of Thai regional cuisine -- finding that ingredient patterns encode centuries of trade routes and cultural boundaries that no existing map shows. The interactive result is live at flavormap.th."

2. What TunDee Already Proves (Skills Not to Repeat)

FlavorMap must demonstrate skills that TunDee does not. An admissions reader who sees two projects using the same techniques is reading about one skillset presented twice. Every technical method in FlavorMap should be genuinely new.

TunDee already covers	Therefore FlavorMap must NOT primarily use
PostgreSQL relational database design	Standard relational DB as the primary data store (use it as supporting infrastructure only)
Web scraping (BeautifulSoup, Playwright, PDFPlumber)	Scraping as the main technical contribution (she scrapes here too, but it's just data collection)
Thai NLP / WangchanBERTa fine-tuning	BERT-based classification as the core ML method
LLM-assisted structured data extraction	Prompt engineering as the primary novelty
Supervised classification (content-based filtering)	Simple classification as the headline method
Collaborative filtering	Recommender systems
Fairness-aware ML (Dwork, Hardt frameworks)	Algorithmic fairness as the research contribution
Causal inference (DID, propensity matching)	Causal methods as the statistical approach
Full-stack web app (Next.js, FastAPI, Supabase)	Platform building as the primary achievement

3. New Technical Skills FlavorMap Adds

Every skill below is genuinely absent from TunDee. An admissions reader comparing the two projects sees a student who has mastered two distinct technical paradigms — not one paradigm applied twice.

Skill	What it is	Where in FlavorMap	Why it matters for DS admissions
Graph theory & network analysis	Representing relationships as nodes and edges; measuring centrality, community structure, clustering coefficients	Ingredient co-occurrence network; community detection for flavor clusters	Graph ML is one of the hottest DS research areas. Used in social networks, drug discovery, recommendation. Rarely seen in high school portfolios.
Unsupervised learning	Finding structure in data without labeled training examples — clustering, dimensionality reduction, anomaly detection	K-means and hierarchical clustering on recipe vectors; UMAP for visualization	TunDee uses only supervised learning. Unsupervised methods show she can explore data without a predefined answer.
Geospatial data science	Working with geographic coordinates, spatial joins, choropleth mapping, spatial autocorrelation (Moran's I)	Mapping ingredient patterns to Thai provinces; spatial autocorrelation of culinary similarity	Geospatial DS is a distinct technical track. PostGIS, Folium, Geopandas are industry-standard tools she demonstrates here.
Interactive data visualization	Building web-based interactive charts and maps users can explore — not static plots but dynamic, queryable visual interfaces	The primary public deliverable — a clickable map of Thai culinary geography at flavormap.th	Visualization is the skill that makes research legible to non-experts. A live demo is the most convincing admissions artifact.

Skill	What it is	Where in FlavorMap	Why it matters for DS admissions
Research design & methodology	Forming a hypothesis, designing a study to test it, selecting appropriate methods, acknowledging limitations honestly	The research paper — structured around a formal hypothesis with a null result possible	TunDee is product-first. FlavorMap is research-first. Having both shows methodological range.
Fieldwork & ethnographic data collection	Structured interviews, informed consent, qualitative coding, integrating qualitative and quantitative data	20–30 cook interviews generating province-tagged recipe data and qualitative context	Mixed-methods research — combining fieldwork with computational analysis — is rare and valued at iSchool programs.
Dataset curation & documentation	Data cards, versioning, bias documentation, HuggingFace dataset release with full metadata	OpenFlavorTH dataset release as public research infrastructure	Releasing a documented public dataset signals she thinks beyond her own work — she is contributing to a research community.
Food image classification (stretch goal)	Fine-tuning EfficientNet or MobileNet on Thai dish photos — computer vision on a domain-specific dataset she collects	Optional visual search feature on the interactive map	Adds computer vision to her portfolio. If built, shows she can work across modalities (tabular + image).

4. The Research Questions

The primary research question: Do ingredient co-occurrence patterns in Thai home cooking vary systematically by province — and if so, do those patterns correlate more strongly with geographic proximity, historical trade routes, or linguistic/ethnic boundaries?

Why this specific question

The question has four properties that make it ideal for a research paper. First, it is falsifiable — there might be no systematic variation, which would itself be an interesting finding. Second, it has multiple possible explanations (geography, trade, language) that she can test against each other. Third, it connects data science to cultural history, making it interesting to multiple audiences. Fourth, nobody has asked this question formally before — the finding is novel regardless of what the data shows.

Five sub-questions (each becomes a section of the paper)

Sub-question	Method	Expected finding
RQ1: What are the most central ingredients in Thai cuisine overall — the ones that appear across all regions?	Network centrality (degree centrality, betweenness centrality) in the full ingredient co-occurrence graph	Garlic, shallots, fish sauce, palm sugar, chilies expected to be high-centrality 'bridge' ingredients. Thai basil will likely show regional variation.
RQ2: Do regional ingredient communities emerge from the network — and do they align with Thailand's four geographic regions?	Louvain community detection on the ingredient network; compare detected communities to North/NE/Central/South regional labels	Expect 4–6 communities. Hypothesis: communities will align better with regions than with administrative provinces.
RQ3: Which provinces have the most distinctive cuisine — the ingredients that appear almost exclusively in their recipes?	TF-IDF adapted for ingredient data: 'ingredient importance' = frequency in province's recipes $\times$ rarity across all provinces	Chiang Mai (Northern herbs), Pattani (Muslim-influenced spices), and Isaan provinces (pla ra, khao niao) expected to be most distinctive.
RQ4: Does culinary similarity between provinces correlate with geographic distance?	Spatial autocorrelation (Moran's I statistic) on ingredient similarity matrix. Mantel test correlating ingredient distance matrix with geographic distance matrix.	Hypothesis: significant positive spatial autocorrelation — adjacent provinces have more similar cuisines. But border provinces (near Laos, Malaysia, Myanmar) may show stronger cross-border similarity than within-Thailand geographic proximity.
RQ5: Can a machine learning model predict a dish's province of origin from its ingredients alone — and where does it fail?	Province-of-origin classifier trained on recipe ingredient vectors. Confusion matrix analysis shows which provinces are most often confused with each other.	Expect ~40–60% accuracy at province level (77 classes is hard). Accuracy at regional level (4 classes) expected ~75–85%. Failure analysis is as interesting as success.

Important: she should be prepared for null results. If ingredient patterns do NOT vary significantly by province, that is a publishable finding — it suggests Thai cuisine is more nationally homogenized than cultural narratives suggest. A null result honestly reported is more admissions-credible than a perfect result that looks fabricated.

5. Dataset Design — What Data She Needs

Target dataset specification

Field	Description	Why needed
recipe_id	Unique identifier	Primary key for all joins
recipe_name_th	Recipe name in Thai	Thai text — links to NLP analysis
recipe_name_en	Recipe name in English (translated or transliterated)	International accessibility; paper tables
province	Thai province of origin (one of 77 provinces)	The geographic variable — most critical field for all analyses
region	One of four: North / Northeast (Isaan) / Central / South	Aggregation level for regional analysis
ingredients[]	List of ingredient names (Thai + English), standardized to canonical form	Core data for network analysis and clustering
ingredient_quantities[]	Quantities where available (many recipes won't have this)	Optional enrichment — do not require; mark null if absent
cooking_method[]	Primary technique(s): stir-fry, curry, soup, raw/salad, grilled, fermented	Second axis of variation beyond ingredients — shows different dimension of cuisine
dish_category	Nam prik / curry / soup / noodle / rice dish / salad (yam) / dessert / snack	Controls for dish type when analyzing ingredient variation
source_type	web_scraped / interview / cookbook / institutional	Critical for bias analysis — source type affects ingredient representation
source_url	URL if scraped; informant ID if interview	Provenance and reproducibility
collection_date	ISO date of collection	Temporal freshness tracking
province_confidence	High / Medium / Low — how confident is the province attribution?	Many recipes will have ambiguous province attribution. This field enables sensitivity analysis.
informant_age_bracket	If interview source: <40 / 40–60 / 60+ years old	Enables analysis of generational recipe variation

Target dataset size

Source	Target recipes	Province coverage	When acquired
Web scraping (Wongnai + SorKorPor + other databases)	1,500	All 77 provinces (uneven)	Months 1–3
Cook interviews (20–30 sessions)	300–500	10–15 provinces (targeted to fill gaps)	Months 2–5
Thai cookbooks (National Library)	200–300	Regional cookbooks fill historical gaps	Months 3–4
Institutional sources (Tourism Authority, SorKorPor)	200–300	Structured province attribution	Month 2

Source	Target recipes	Province coverage	When acquired
TOTAL TARGET	2,200–2,600	All 77 provinces; min. 20 recipes/province for reliable analysis	Month 5

The minimum viable dataset for meaningful analysis: 20 recipes per province $\times$ 77 provinces = 1,540 recipes with confident province attribution. Everything above this is enrichment. She should aim for this minimum before running any network analysis.

6. Data Source 1 — Web Scraping Thai Recipe Databases

Primary scraping targets

Source	URL	Recipes est.	Province attribution	Scraping approach
Wongnai Recipe	recipe.wongnai.com	50,000+	Tag-based; inconsistent. Users tag region, not always province.	Playwright (JavaScript-rendered). Filter by region tags. Expect 30–40% usable province attribution.
SorKorPor (กรมส่งเสริมการเกษตร)	www.doa.go.th	2,000+	Excellent — organized by province explicitly. Highest quality source.	BeautifulSoup (static HTML). Priority target — structure is clean.
Thai Taste (ไทยรสแซ่บ)	thaifoodrecipe.com	3,000+	Regional tags available. Province-level inconsistent.	BeautifulSoup. Extract region tags; LLM infer province from context.
Kapook Cooking	cooking.kapook.com	10,000+	Category tags (Northern, Isaan etc.) but not province.	BeautifulSoup. Useful for regional-level data.
Tourism Authority of Thailand	amazingthailand.com	500+	Excellent province attribution — TAT organizes by destination.	BeautifulSoup. Small but high quality — strong province attribution.
Pantip Food Board	pantip.com/club/food	Large but messy	Community posts — province mention often in text body.	NLP extraction of province mentions from post text. Lower reliability.

Ingredient standardization — the hardest scraping problem

The same ingredient appears under dozens of different names across Thai recipe sources. This is the most time-consuming data cleaning task and must be solved before any network analysis is possible.

- **The problem in concrete terms:** 'กระเทียม' (garlic), 'กระเทียมไทย' (Thai garlic), 'กระเทียมพ่น' (sliced garlic), 'กระเทียมบด' (minced garlic), 'กระเทียมสับ' (chopped garlic) — all the same ingredient for FlavorMap's purposes but will appear as five different nodes in the network if not standardized.
- **The solution:** Build a canonical ingredient dictionary mapping every variant name to a standardized form. Start with 200 most common ingredients manually (she + ISB Club team). Then use WangchanBERTa semantic similarity to auto-map variants to canonical forms for the long tail.
- **The hierarchy:** Ingredients should be standardized at the right level of specificity. 'Fish sauce' is one ingredient. 'Pla ra' (fermented freshwater fish) is a different ingredient from 'nam pla' (fish sauce from saltwater fish) — this distinction matters for Isaan vs. Central Thai analysis. Do NOT over-aggregate.
- **Quantity standardization:** If quantities are available, standardize to grams. If only 'a handful' or 'to taste' — record the presence but not the quantity. Do not impute fictional quantities.
- **The canonical ingredient list should be released as part of the OpenFlavorTH dataset** — it is itself a valuable research contribution.

Province attribution — the second hardest scraping problem

Most online recipes are not tagged with a specific province. They may be tagged with a region ('Isaan style'), a dish category ('Northern curry'), or nothing at all. She needs a strategy for attributing province to recipes that lack explicit tags.

- **Tier 1 — Explicit province tag:** Recipe says 'จากจังหวัดสุรินทร์' (from Surin province). Use directly. ~15–20% of scraped recipes.
- **Tier 2 — Explicit region tag:** Recipe says 'อาหารอีสาน' (Isaan food). Assign to region, not province. Include in regional analysis; exclude from province-level analysis unless enriched by interview.

- **Tier 3 — Dish name inference:** Some dish names encode province. 'ข้าวมันไก่ตรัง' (Trang-style chicken rice) = Trang province. Build a lookup dictionary of dish name ↔ province mappings for the 200 most common province-encoded dish names.
- **Tier 4 — LLM inference:** For ambiguous cases, use Claude API with a structured prompt: 'Based on this recipe's ingredients and context, which Thai province is most likely the origin? Output JSON with province, confidence (high/medium/low), and reasoning.' Use only high-confidence LLM attributions; mark medium/low as province_confidence='medium'/'low' in the database.
- **Never fabricate:** A recipe with no province attribution stays province=null. A 'low confidence' attribution gets included in sensitivity analysis but not the main analysis. Intellectual honesty about data quality is what makes the paper defensible.

Sample scraping code structure (Wongnai)

```
# Pseudocode structure – actual implementation with coding mentor from playwright.async_api import
async_playwright import asyncio, json async def scrape_wongnai_recipe(url): async with
async_playwright() as p: browser = await p.chromium.launch() page = await browser.new_page() await
page.goto(url) await page.wait_for_selector('.recipe-detail') name = await
page.text_content('h1.recipe-title') region_tags = await page.query_selector_all('.tag-region')
ingredient_blocks = await page.query_selector_all('.ingredient-item') ingredients = [] for block
in ingredient_blocks: name_th = await block.text_content('.ingredient-name') quantity = await
block.text_content('.ingredient-qty') ingredients.append({'name_th': name_th, 'quantity':
quantity}) return { 'recipe_name_th': name, 'region_tags': [await t.text_content() for t in
region_tags], 'ingredients': ingredients, 'source_url': url, 'source_type': 'web_scraped',
'collection_date': datetime.date.today().isoformat() }
```

7. Data Source 2 — Fieldwork Interviews with Home Cooks

Why fieldwork is non-negotiable. Web-scraped recipes are biased toward urban, educated, and digitally active cooks. The most distinctive provincial recipes — the ones encoded with the most local knowledge — exist only in the kitchens of elderly home cooks who never posted them online. These recipes are the most important data for her research question AND the most powerful source of human stories for the essay. She cannot skip this.

Who to interview

Target: 20–30 home cook interviews. Priority characteristics:

- **Age 55+** — carries pre-internet, pre-modernization cooking knowledge
- **Born and raised in one province** — authentic provincial tradition, not Bangkok-adapted version
- **Does not have a social media presence** — if they post online, their recipes are already in the web data
- **Cooks from memory, not recipes** — the goal is tacit knowledge that has never been written down
- **Provincial distribution:** prioritize provinces with fewer than 20 web-scraped recipes to fill coverage gaps

How to find interviewees

- **ISB network:** Ask ISB staff, teachers, and parents — many have relatives from different provinces who cook. This is the fastest starting point.
- **Dad's Chulalongkorn network:** Chulalongkorn researchers and administrative staff come from provinces across Thailand. Dad introduces her as a research student studying Thai culinary heritage.
- **EEF provincial contacts:** EEF has coordinators in every province for TunDee. The same contacts who help with TunDee can connect her to home cooks in their communities.
- **Weekend markets in Bangkok:** Or Tor Kor Market (อ.ต.ก.), Chatuchak Weekend Market, and Bangkok's community markets have vendors from specific provinces who cook traditional dishes. She approaches them directly, explains the research.
- **Thai community centers by region:** Bangkok has community associations for each Thai region (สมาคมชาวอีสาน, สมาคมชาวเหนือ). These associations actively welcome cultural preservation research.

Interview protocol

Stage	Duration	What happens	Data collected
Pre-interview consent	5 min	Explain the research purpose. Obtain verbal informed consent for recording and use in research. Explain the data will be anonymized and used only for research purposes. Mom present as chaperone and witness.	Signed or verbal consent recorded
Background	10 min	Province and district of birth/childhood. Years in Bangkok if migrated. Who taught them to cook (mother, grandmother, specific person). Which dishes they consider 'their' traditional recipes.	Informant profile: province, age bracket, learning source
Recipe elicitation	30–45 min	Ask them to describe 3–5 dishes they consider uniquely from their province. For each dish: list all ingredients from memory (do NOT show them a written recipe), describe the cooking method, explain what makes it different from the Bangkok or restaurant version. Video record with permission.	Ingredient lists per dish with province attribution (high confidence). Qualitative notes on distinctiveness.

Stage	Duration	What happens	Data collected
Validation	10 min	Show them 2–3 scraped recipes from their province. Do they recognize them? What is missing or wrong? This validates the web-scraped data against expert knowledge.	Quality assessment of scraped data for their province
Photo documentation	5 min	Photograph key ingredients they mention (if visible in their kitchen or market). These photos become training data for the food image classifier stretch goal.	Province-labeled food photos for image classifier

Fieldwork logistics and safety

- **Mom is always present** as chaperone for all fieldwork sessions. Non-negotiable for a 15-year-old researcher.
- **Interviews conducted in Thai.** She leads in Thai. Mom provides additional assurance for hesitant interviewees.
- **Recording protocol:** Phone recording with explicit verbal consent. Keep all recordings until the paper is published, then delete as promised.
- **Anonymization:** Each informant is assigned an ID (e.g., INT_CM_001 for Chiang Mai). Their name is never used in the paper or dataset.
- **Compensation:** A small gift (fruit, Thai sweets, 200–300 THB equivalent) as a gesture of appreciation. Not framed as payment for information.
- **IRB-equivalent approval:** For the research paper to be publishable, the interview methodology should be reviewed by Chulalongkorn's or Mahidol's research ethics board (IRB equivalent). Dad facilitates this through his faculty contacts. This is a one-page form and takes 2–4 weeks.

Provincial coverage strategy — who to prioritize

Province / Region	Why priority	Accessibility from Bangkok
Chiang Mai (North)	Most distinctive Northern cuisine. Large diaspora in Bangkok. Khantoke dining tradition well-documented but home cooking variants rare.	1 hr flight or Bangkok Chiang Mai community (no travel needed)
Surin / Buriram (NE Isaan)	TunDee's primary user base. EEF contacts already established. Pla ra (fermented fish) cuisine most distinct from Central Thai.	5–6 hr bus or overnight train. Or Bangkok Isaan community.
Songkhla / Pattani (South)	Southern Thai cuisine most distinct due to Muslim influence, coconut-heavy curries. Underrepresented in online databases.	1 hr flight. Southern Thai community associations in Bangkok.
Samut Songkhram (Central)	Traditional Central Thai cuisine hub. Near Bangkok (90 min). Strong floating market and traditional food culture.	90 min drive. Easiest fieldwork target.
Bangkok diaspora cooks (from all provinces)	Elderly migrants who moved to Bangkok 20–30 years ago but still cook provincial recipes from memory. Most time-efficient fieldwork approach.	No travel. Accessible through community associations in Bangkok.

Practical advice: Bangkok-based diaspora interviews first (Months 2–3), provincial field trips only if needed to fill specific coverage gaps (Months 4–5). A student in Bangkok can interview 20 people from 15 provinces without leaving the city by connecting with community associations. Travel adds fieldwork richness but is not required for minimum viable dataset.

Step	What happens	Tool	Hours est.
1. Raw data ingestion	All scraped and interview data loaded into PostgreSQL. Raw table stores original text exactly as collected — never overwrite raw data.	Python + PostgreSQL	5 hrs
2. Duplicate detection	Same recipe from multiple sources. Use recipe name + ingredient list similarity (Jaccard similarity > 0.85) to flag duplicates. Manual review of flagged pairs. Keep highest-quality source.	Python (fuzzywuzzy, difflib)	8 hrs
3. Ingredient extraction from Thai text	For scraped recipes, ingredients are often embedded in Thai prose ('ใส่กระเทียม 3 กลีบ'). Use Claude API to extract structured ingredient list from unstructured Thai text. Manual review.	Claude API + manual review	10 hrs
4. Ingredient canonicalization	Map every extracted ingredient to canonical_ingredient_id using the dictionary. Flag unmapped ingredients for manual review (new canonical entry needed).	Python + WangchanBERTa cosine similarity	20 hrs
5. Province attribution	Apply the 4-tier attribution strategy. Tag each recipe with province (if known), region, province_confidence.	Python + Claude API (for tier 4 LLM attribution)	15 hrs
6. Quality flagging	Flag recipes: very short ingredient lists (<3 ingredients, likely incomplete); very long lists (>25 ingredients, likely composite/fusion); missing province with no recovery possible.	Python rule-based filters	5 hrs
7. Final cleaned table	recipes_clean table with all quality flags. Research analysis uses only province_confidence IN ('high', 'medium') and ingredient_count BETWEEN 3 AND 25.	SQL view	3 hrs

Step	What happens	Tool	Hours est.
TOTAL CLEANING			~66 hrs

9. Technical Layer 1 — Ingredient Network Analysis

This is the core technical contribution. The ingredient co-occurrence network is what makes FlavorMap genuinely different from a recipe database or a food visualization. It encodes relationships between ingredients that no flat database can represent. Every admissions reader who asks 'what did you build?' should get this answer first.

What an ingredient co-occurrence network is

A graph where: each node is a canonical ingredient. An edge between ingredient A and B exists if they appear together in at least one recipe. The edge weight is the number of recipes in which they co-occur. Heavier edges = stronger culinary association. The structure of this graph encodes the 'grammar' of Thai cooking.

Step-by-step implementation

Step 1 — Build the co-occurrence matrix

```
# Build ingredient co-occurrence matrix from recipe database
import pandas as pd
import numpy as np
from itertools import combinations
from collections import Counter
import networkx as nx

# Load recipes with ingredient lists
recipes = pd.read_sql( "SELECT recipe_id, province, region,
array_agg(canonical_id) as ingredients " "FROM recipe_ingredients ri " "JOIN recipes r USING
(recipe_id) " "WHERE province_confidence IN ('high', 'medium') " "GROUP BY recipe_id, province,
region", conn )

# Build co-occurrence counter
cooccurrence = Counter()

for _, row in recipes.iterrows():
    ings = list(set(row['ingredients']))
    # deduplicate within recipe
    for pair in combinations(sorted(ings), 2):
        cooccurrence[pair] += 1

# Build NetworkX graph - weighted, undirected
G_full = nx.Graph()

for (a, b), weight in cooccurrence.items():
    if weight >= 3:
        # minimum 3 co-occurrences to include edge
        G_full.add_edge(a, b, weight=weight)

print(f"Nodes: {G_full.number_of_nodes()}, Edges: {G_full.number_of_edges()}")
```

Step 2 — Build regional sub-networks

Build four separate networks — one per region (North, Northeast, Central, South) — using only recipes from that region. Compare the structure of these four networks to identify regional differences. Which ingredient pairs are strongly associated in Isaan but rarely appear together in Southern cuisine?

```
# Build per-region networks
regions = ['North', 'Northeast', 'Central', 'South']
G_regional = {}

for region in regions:
    region_recipes = recipes[recipes['region'] == region]
    cooc = Counter()

    for _, row in region_recipes.iterrows():
        ings = list(set(row['ingredients']))
        for pair in combinations(sorted(ings), 2):
            cooc[pair] += 1

    G = nx.Graph()
    for (a, b), weight in cooc.items():
        if weight >= 2:
            # lower threshold for smaller regional datasets
            G.add_edge(a, b, weight=weight)

    G_regional[region] = G
```

Step 3 — Network centrality analysis (answers RQ1)

- **Degree centrality:** Which ingredients appear with the most other ingredients? High degree = versatile, cross-regional ingredient.
- **Betweenness centrality:** Which ingredients act as 'bridges' between otherwise distinct flavor clusters? High betweenness = culturally connective ingredient.
- **PageRank:** Which ingredients are connected to other high-importance ingredients? PageRank captures 'prestige' rather than raw connectivity.
- **Eigenvector centrality:** Which ingredients cluster with other important ingredients? Reveals the 'core' of Thai cooking vs. the periphery.

```
# Compute centrality metrics for each ingredient
centrality_metrics = pd.DataFrame({ 'ingredient':
list(G_full.nodes()), 'degree_centrality': dict(nx.degree_centrality(G_full)).values(),
'betweenness_centrality': dict( nx.betweenness_centrality(G_full, weight='weight')).values(),
'pagerank': dict(nx.pagerank(G_full, weight='weight')).values(), 'eigenvector_centrality': dict(
nx.eigenvector_centrality(G_full, weight='weight')).values(), })

# Join with ingredient metadata
(name_th, name_en, regional_significance)
centrality_metrics = centrality_metrics.merge(
canonical_ingredients_df, on='ingredient')

centrality_metrics.sort_values('betweenness_centrality', ascending=False).head(20)
```

Step 4 — Community detection (answers RQ2)

Community detection finds groups of ingredients that appear together more often than expected by chance. These communities represent 'flavor clusters' — the fundamental building blocks of Thai cuisine. The research question: do these algorithmically detected communities align with regional boundaries?

```
# Louvain community detection import community as community_louvain # pip install python-louvain
partition = community_louvain.best_partition(G_full, weight='weight') # partition =
{ingredient_id: community_number, ...} # Evaluate: do communities align with regions? # For each
community, what fraction of recipes containing those ingredients are from each region?
community_df = pd.DataFrame([ {'ingredient': ing, 'community': comm} for ing, comm in
partition.items() ]) # Compare community membership to regional ingredient profiles # Use
Normalized Mutual Information (NMI) or Adjusted Rand Index (ARI) # to measure alignment between
detected communities and ground-truth regions
```

Step 5 — Region-specific ingredient distinctiveness (answers RQ3)

Adapted TF-IDF for ingredients: an ingredient is 'distinctive' to a province if it appears frequently in that province's recipes (high TF) and rarely across all provinces (high IDF). This produces a ranked list of the most province-defining ingredients for each of the 77 provinces.

```
# TF-IDF adapted for ingredient-province distinctiveness from sklearn.feature_extraction.text
import TfidfVectorizer # Represent each province as a 'document' of ingredient lists province_docs
= recipes.groupby('province').apply( lambda x: ' '.join([ ' '.join(row['ingredients']) for _, row
in x.iterrows() ]) ).reset_index() province_docs.columns = ['province', 'ingredient_text'] #
TF-IDF vectorization tfidf = TfidfVectorizer(tokenizer=lambda x: x.split(), lowercase=False)
tfidf_matrix = tfidf.fit_transform(province_docs['ingredient_text']) # tfidf_matrix[i, j] =
distinctiveness of ingredient j for province i feature_names = tfidf.get_feature_names_out() # Top
10 most distinctive ingredients per province for i, province in
enumerate(province_docs['province']): top_indices =
tfidf_matrix[i].toarray()[0].argsort()[-10:][::-1] top_ingredients = [(feature_names[j],
tfidf_matrix[i, j]) for j in top_indices] print(f"{province}: {top_ingredients}")
```

10. Technical Layer 2 — Geographic Clustering

Representing each province as a culinary vector

Before geographic clustering is possible, each province must be represented as a numerical vector encoding its culinary profile. The most natural representation: a vector of length N (number of canonical ingredients) where each component is the TF-IDF score of that ingredient for that province.

Dimensionality reduction with UMAP

77 provinces × 400+ ingredients is a high-dimensional space. UMAP (Uniform Manifold Approximation and Projection) reduces this to 2D for visualization while preserving local structure. Unlike PCA (which she should also run for comparison), UMAP preserves non-linear relationships and is better suited for ingredient data where the interesting structure is in clusters, not in linear components.

```
# UMAP dimensionality reduction and clustering
import umap from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.metrics import silhouette_score
import geopandas as gpd
import folium # Step 1: UMAP reduction for visualization
reducer = umap.UMAP(n_components=2, n_neighbors=10, min_dist=0.1, metric='cosine', random_state=42)
province_2d = reducer.fit_transform(tfidf_matrix.toarray())
# Step 2: Clustering in original high-dimensional space
# Try K-means with k=4 (regions) and k=6,8 (sub-regional clusters)
for k in [4, 6, 8]:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=20)
    labels = kmeans.fit_predict(tfidf_matrix.toarray())
    sil_score = silhouette_score(tfidf_matrix.toarray(), labels)
    print(f"K={k}: silhouette score = {sil_score:.3f}")
# Step 3: Hierarchical clustering (more interpretable for the paper)
hclust = AgglomerativeClustering(n_clusters=None, distance_threshold=0.5, linkage='ward')
hclust_labels = hclust.fit_predict(tfidf_matrix.toarray())
# Dendrogram shows which provinces are most similar – useful for paper figure
```

Spatial autocorrelation analysis — Moran's I

The key geographic analysis: do culinarily similar provinces tend to be geographically adjacent? Moran's I statistic measures spatial autocorrelation — whether values (in this case, culinary similarity) are more clustered in space than would be expected by chance. A significant positive Moran's I means 'yes, similar cuisines cluster geographically.'

```
# Moran's I spatial autocorrelation
import libpysal from esda.moran import Moran
import geopandas as gpd
# Load Thailand province shapefile (available from Thailand GADM)
thai_map = gpd.read_file('thailand_provinces.shp')
# Merge culinary cluster labels with geographic data
thai_map = thai_map.merge(pd.DataFrame({'province': province_docs['province'], 'culinary_cluster': kmeans_labels, 'umap_dim1': province_2d[:, 0], 'umap_dim2': province_2d[:, 1]}), on='province')
# Build spatial weights matrix (queen contiguity – shares border or corner)
w = libpysal.weights.Queen.from_dataframe(thai_map)
w.transform = 'r' # row-standardize
# Moran's I for culinary cluster labels
moran = Moran(thai_map['umap_dim1'], w)
print(f"Moran's I = {moran.I:.4f}, p-value = {moran.p_sim:.4f}")
# If p < 0.05: geographic clustering is statistically significant
```

Mantel test — correlating culinary distance with geographic distance

The Mantel test asks: is culinary distance (difference in ingredient profiles) correlated with geographic distance (kilometers between province centroids)? A significant positive Mantel correlation means 'the further apart two provinces are, the more different their cuisine' — the most direct test of geography-cuisine correlation.

```
# Mantel test from scipy.spatial.distance
import pdist, squareform from scipy.stats import pearsonr
import numpy as np
# Culinary distance matrix (cosine distance between province ingredient vectors)
culinary_dist = squareform(pdist(tfidf_matrix.toarray(), metric='cosine'))
# Geographic distance matrix (haversine distance between province centroids)
from geopy.distance import great_circle
centroids = thai_map[['province', 'geometry']].copy()
centroids['centroid'] = centroids['geometry'].centroid
centroids['lat'] = centroids['centroid'].y
centroids['lon'] = centroids['centroid'].x
n = len(centroids)
geo_dist = np.zeros((n, n))
for i in range(n):
    for j in range(n):
        geo_dist[i, j] = great_circle(centroids.iloc[i]['lat'], centroids.iloc[i]['lon'], centroids.iloc[j]['lat'], centroids.iloc[j]['lon']).km
# Mantel test: correlate upper triangles
upper = np.triu_indices(n, k=1)
r, p = pearsonr(culinary_dist[upper], geo_dist[upper])
print(f"Mantel r = {r:.4f}, p = {p:.4f}")
```

11. Technical Layer 3 — Province-of-Origin Classifier

Why build a classifier

The classifier answers RQ5: can a machine predict a dish's province of origin from its ingredients? This is a concrete, demonstrable result that non-technical audiences (admissions readers, journalists, foundations) can understand intuitively. If the model predicts Chiang Mai from a list of ingredients including galangal, dried chili, and fermented soybean — that is a vivid demonstration that geographic signal genuinely exists in ingredient data.

Classification at two levels

- **Region-level (4 classes):** North, Northeast, Central, South. Easier problem. Expected accuracy 70–85%. This is the headline result.
- **Province-level (77 classes):** Much harder. Expected accuracy 35–55%. The value here is the confusion matrix — which provinces get confused with which? This reveals culinary similarity structure.

Implementation

```
# Province-of-origin classifier from sklearn.ensemble import RandomForestClassifier,
GradientBoostingClassifier from sklearn.linear_model import LogisticRegression from
sklearn.model_selection import StratifiedKFold, cross_val_score from sklearn.metrics import
classification_report, confusion_matrix from sklearn.preprocessing import MultiLabelBinarizer
import matplotlib.pyplot as plt import seaborn as sns # Feature representation: multi-hot encoding
of ingredients mlb = MultiLabelBinarizer() X = mlb.fit_transform(recipes['ingredients']) # shape:
(n_recipes, n_ingredients) y_region = recipes['region'] y_province = recipes['province'] #
Stratified K-fold to handle class imbalance (some provinces have few recipes) skf =
StratifiedKFold(n_splits=5, shuffle=True, random_state=42) # Compare three classifiers classifiers
= { 'Logistic Regression': LogisticRegression(max_iter=1000, C=1.0), 'Random Forest':
RandomForestClassifier(n_estimators=200, random_state=42), 'Gradient Boosting':
GradientBoostingClassifier(n_estimators=100) } for name, clf in classifiers.items(): scores =
cross_val_score(clf, X, y_region, cv=skf, scoring='accuracy') print(f"{name} region accuracy:
{scores.mean():.3f} ± {scores.std():.3f}") # Train final model and produce confusion matrix
best_clf = RandomForestClassifier(n_estimators=200, random_state=42) best_clf.fit(X, y_region) #
retrain on full dataset for feature importance # Feature importance — which ingredients are most
discriminative? feature_importance = pd.DataFrame({ 'ingredient': mlb.classes_, 'importance':
best_clf.feature_importances_ }).sort_values('importance', ascending=False) print("Top 20 most
discriminative ingredients:") print(feature_importance.head(20)) # Confusion matrix for
province-level classifier y_pred_province = cross_val_predict(
RandomForestClassifier(n_estimators=200, random_state=42), X, y_province, cv=skf) cm =
confusion_matrix(y_province, y_pred_province) # The confusion matrix shows which provinces are
most often confused with each other # High off-diagonal values between two provinces = culinarily
similar
```

The confusion matrix as a finding

The most interesting output of the province classifier is not the accuracy — it's the confusion matrix. When the model confuses Province A with Province B, it means the two provinces have similar ingredient profiles. This creates a 'culinary similarity map' that the geographic clustering analysis can validate. If the classifier consistently confuses Surin with Sisaket (adjacent Isaan provinces) but never confuses Surin with Phuket (geographically distant Southern province) — that is a quantitative finding about Thai culinary geography.

Feature importance — the interpretable finding

Random Forest feature importance shows which ingredients are the strongest predictors of regional origin. These should be the same ingredients that the network analysis identified as regionally distinctive (the TF-IDF analysis). Agreement between two independent methods — network analysis AND classifier feature importance — is the strongest evidence that the findings are real, not artifacts.

12. Technical Layer 4 — Interactive Web Visualization

The visualization is the project's public face. A live URL at flavormap.th that any admissions reader can open and explore is worth more than any number of static figures in a PDF. The goal: someone who has never heard of FlavorMap can open it, click on Chiang Mai, and within 30 seconds understand what makes Northern Thai cuisine distinctive.

Three views — what the visualization shows

View 1 — The Culinary Geography Map (landing page)

- A choropleth map of Thailand's 77 provinces, colored by culinary cluster membership
- Hover over any province: see its top 5 most distinctive ingredients (TF-IDF) and its culinary similarity score to its neighbors
- Click any province: zoom in to see all recipes in the database from that province
- Toggle between 4-cluster (regional), 6-cluster (sub-regional), and continuous similarity coloring
- A legend showing which clusters emerged from the analysis (not pre-defined regions)

View 2 — The Ingredient Network (the technical centerpiece)

- Force-directed network graph showing ingredient co-occurrence relationships
- Node size = degree centrality (most-used ingredients are largest nodes)
- Edge thickness = co-occurrence frequency (thicker = appear together more often)
- Node color = Louvain community membership
- Hover over any ingredient: see which provinces use it most frequently
- Filter by region: show only the ingredient network for one region at a time
- Highlight any ingredient: show all its co-occurrence relationships in isolation

View 3 — The Provincial Comparison Tool

- Select two provinces and see a side-by-side comparison of their most distinctive ingredients
- Venn diagram showing shared vs. province-specific ingredients
- Culinary distance score (0 = identical, 1 = completely different)
- Sample recipes from each province for context

Technology stack for visualization

Component	Technology	Why this choice
Map visualization	Folium (Python) for prototyping □ Mapbox GL JS for production	Folium is Python-native and fast to prototype. Mapbox gives smooth interactive map with custom styling. Both use GeoJSON Thailand province boundaries.
Network graph	Sigma.js or Cytoscape.js (JavaScript library)	Both are purpose-built for interactive network visualization in browser. Sigma.js handles larger networks more smoothly. React wrapper available for both.
Frontend framework	Next.js (same as TunDee)	She already knows it. Consistent tech across projects.
Data API	FastAPI (same as TunDee)	Serves ingredient data, province profiles, similarity scores to the frontend.
Hosting	Vercel (frontend) + Railway (FastAPI backend)	Free tier handles traffic volumes expected. Same hosting as TunDee.
Domain	flavormap.th or flavormap.org (\$6.88/yr on Porkbun)	flavormap.th requires Thai company registration (use KrueChai Foundation). flavormap.org is immediately available.

Component	Technology	Why this choice
Thailand GeoJSON	Thailand GADM province boundaries (free, public domain)	Download from gadm.org. Province-level administrative boundaries for all 77 provinces.

Sample visualization code — choropleth map with Folium

```
# Prototype choropleth map – province culinary clusters
import folium
import geopandas as gpd
import json

# Load Thailand province boundaries
thai_geo = gpd.read_file('thai_provinces_gadm.shp')

# Merge with analysis results
thai_geo = thai_geo.merge(
    province_analysis[['province', 'culinary_cluster', 'top_ingredient_1', 'top_ingredient_2', 'top_ingredient_3']],
    left_on='NAME_1', right_on='province'
)

# Color palette: one color per culinary cluster
cluster_colors = {0: '#1a6e3d', 1: '#2a5ca8', 2: '#8b2c2c', 3: '#a87a1c', 4: '#0f5e56', 5: '#4a1a6e'}

# Create Folium map centered on Thailand
m = folium.Map(location=[13.0, 101.5], zoom_start=6, tiles='CartoDB positron')

# Add choropleth layer
folium.Choropleth(
    geo_data=thai_geo.to_json(),
    data=province_analysis,
    columns=['province', 'culinary_cluster'],
    key_on='feature.properties.province',
    fill_color='YlOrRd',
    fill_opacity=0.7,
    line_opacity=0.2,
    legend_name='Culinary Cluster'
).add_to(m)

# Add hover tooltips
folium.features.GeoJson(
    thai_geo.to_json(),
    tooltip=folium.features.GeoJsonTooltip(
        fields=['province', 'top_ingredient_1', 'top_ingredient_2'],
        aliases=['Province:', 'Most distinctive ingredient:', '2nd most distinctive:']
    )
).add_to(m)

m.save('flavormap_prototype.html')
```

13. Technical Layer 5 — Food Image Recognition (Stretch Goal)

This is a stretch goal — attempt only if ahead of schedule by Month 7. The four core layers (network analysis, geographic clustering, classifier, visualization) are sufficient for a strong paper and strong admissions story. The image classifier adds computer vision skills but should not delay the core work.

What it adds if built

- **Computer vision skills** — EfficientNet/MobileNet fine-tuning is a genuinely new skill not in TunDee
- **A visual search feature** — users can photograph a Thai dish and see which province it most likely comes from
- **A multimodal dataset contribution** — province-labeled Thai food photos are a rare and valuable open dataset
- **A richer paper** — comparing ingredient-based classification accuracy to image-based classification accuracy is an interesting experimental design

Implementation overview

```
# Food image classifier — EfficientNet fine-tuning
import torch
import torchvision.transforms as transforms
from torchvision.models import efficientnet_b0
from torch.utils.data import Dataset, DataLoader
from PIL import Image
import os

# Dataset: 200+ photos per region collected during fieldwork
# Minimum viable: 50 photos per region × 4 regions = 200 photos
# Strong: 100 photos per province × 20 provinces = 2,000 photos
class ThaiDishDataset(Dataset):
    def __init__(self, image_dir, labels_csv, transform=None):
        self.data = pd.read_csv(labels_csv)
        # columns: image_path, region, province
        self.image_dir = image_dir
        self.transform = transform or transforms.Compose([
            transforms.Resize((224, 224)),
            transforms.ToTensor(),
            transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])
        ])
        # Load pre-trained EfficientNet-B0 and fine-tune on Thai dish photos
        model = efficientnet_b0(pretrained=True)
        num_classes = 4 # regions (or more for province-level)
        model.classifier[1] = torch.nn.Linear(model.classifier[1].in_features, num_classes)
        # Training: freeze early layers, fine-tune last 3 blocks + classifier
        for name, param in model.named_parameters():
            if 'features.7' not in name and 'features.8' not in name and 'classifier' not in name:
                param.requires_grad = False
```

14. Full Technology Stack

Category	Technology	Use in FlavorMap	New vs. TunDee
Data storage	PostgreSQL + PostGIS extension	Recipe database + spatial queries on province geography	PostGIS spatial extension is NEW
Web scraping	BeautifulSoup + Playwright + PDFPlumber	Scraping recipe databases (same tools as TunDee)	Same as TunDee
Thai NLP	WangchanBERTa + Claude API	Ingredient name standardization; province inference from text	Same as TunDee (reuse)
Graph analysis	NetworkX + python-louvain + igraph	Ingredient co-occurrence network; community detection; centrality metrics	ENTIRELY NEW — not in TunDee
Unsupervised ML	scikit-learn (KMeans, AgglomerativeClustering) + UMAP	Province clustering by ingredient profile; dimensionality reduction	ENTIRELY NEW — not in TunDee
Geospatial	GeoPandas + libpysal + Folium + Mapbox GL JS	Province boundary mapping; Moran's I; choropleth visualization	ENTIRELY NEW — not in TunDee
Statistical analysis	scipy + pingouin (for Mantel test)	Spatial autocorrelation; Mantel test; statistical significance testing	Partially new (TunDee uses statsmodels for causal inference)
Supervised ML	scikit-learn (RandomForest, GradientBoosting, LogisticRegression)	Province-of-origin classifier; feature importance	Same library, different application
Interactive viz	Sigma.js or Cytoscape.js + Mapbox GL JS + Plotly	Network graph visualization + choropleth map + comparison charts	ENTIRELY NEW — not in TunDee
Computer vision (stretch)	PyTorch + EfficientNet-B0 + torchvision	Food image classifier for province-of-origin prediction	ENTIRELY NEW — not in TunDee
Frontend	Next.js + Tailwind CSS	flavormap.th web interface	Same as TunDee
Backend API	FastAPI (Python)	Serves ingredient data, network data, map data to frontend	Same as TunDee
Dataset release	HuggingFace Datasets + data card documentation	OpenFlavorTH public dataset release	NEW — no TunDee dataset released publicly
Hosting	Vercel + Railway + PostgreSQL (Supabase)	Same infrastructure as TunDee — no new accounts needed	Same as TunDee

Infrastructure cost estimate: \$0–30/month (free tiers for hosting, same Supabase as TunDee). GPU cost for image classifier if built: ~\$30–50 one-time on Google Colab. Domain: \$6.88/yr for flavormap.org on Porkbun.

15. The Research Paper — Plan and Structure

The paper is the primary academic output of FlavorMap. The visualization is the public-facing output. Both matter for admissions. But the paper is what tells an admissions committee that she operates at the level of real research, not just real products.

Paper identity

Element	Detail
Proposed title	"Ingredient networks as cultural geography: a computational analysis of Thai regional cuisine variation across 77 provinces"
Authors	Her (first author) · Chulalongkorn food science or cultural studies faculty (senior author) · optionally, a digital humanities or computational linguistics researcher as second author
Target word count	8,000–10,000 words (conference paper length)
Primary submission target	Digital Humanities 2026 (DH2026) — July 2026 submission deadline. The most appropriate venue: explicitly values interdisciplinary computational approaches to cultural questions.
Secondary targets	Journal of Cultural Analytics (open access, peer-reviewed); ACM CHI 2027 (cultural heritage computing track); ISMIR 2026 Late Breaking Demo (if image classifier built)
Fallback	arXiv preprint (cs.SI — Social and Information Networks, or cs.CY — Computers and Society). Posted in Month 11 regardless of conference acceptance status.
Language	English (for international venues). Thai summary for Thai academic conferences.
Open access	All data, code, and paper will be fully open — GitHub + HuggingFace + arXiv. Transparency is non-negotiable for credibility.

Full paper outline — section by section

Abstract (250 words)

One sentence each: motivation, data (2,200+ Thai recipes across 77 provinces), methods (network analysis + geographic clustering + classification), key finding, implication. The abstract is the most-read part of any paper — write it last but place it first.

1. Introduction (1,000 words)

The cultural geography of food. Thai cuisine as a unified category vs. the reality of regional variation. The gap: no prior computational analysis exists. Her contribution: first province-level ingredient network analysis. Structure of the paper. Her research questions (RQ1–RQ5) stated explicitly.

2. Related Work (800 words)

Four bodies of literature: (a) flavor networks and computational food science (Ahn et al. 2011, West et al. 2013 — regional cuisine paper); (b) digital humanities approaches to food culture; (c) geospatial analysis of cultural practices; (d) Thai culinary scholarship (academic ethnography of Thai food). This section proves she has read the field. It should include 15–20 citations.

3. Data Collection (1,000 words)

Web scraping methodology (sources, volumes, scraping approach). Interview methodology (informant selection, interview protocol, consent procedures, provincial coverage). Data cleaning pipeline. Canonical ingredient dictionary. Province attribution strategy and confidence tiers. Honest discussion of data limitations and biases.

4. Methods (1,500 words)

Four subsections: (4.1) Ingredient network construction and centrality analysis. (4.2) Community detection and regional alignment. (4.3) Geographic clustering — UMAP, K-means, hierarchical clustering, Moran's I, Mantel test. (4.4) Province-of-origin classifier. Each method explained at the level of 'a data scientist could reproduce this from this description.'

5. Results (2,000 words)

One subsection per research question. RQ1: centrality findings — which ingredients are the most central in Thai cuisine? RQ2: do algorithmically detected communities align with regional boundaries? RQ3: which provinces have the most distinctive cuisines? RQ4: Moran's I and Mantel test results — is geographic proximity correlated with culinary similarity? RQ5: classifier accuracy at region and province levels; confusion matrix analysis; feature importance findings.

6. Discussion (1,500 words)

What do the findings mean? Which findings confirm intuitions about Thai food? Which findings are surprising? The connection to cultural history: which detected patterns can be explained by trade routes, migration, colonial history, ecological constraints? Comparison to prior work (Ahn et al., West et al.) — how does Thai cuisine compare to Western and East Asian flavor network structure?

7. Limitations (500 words)

Data biases: urban-educated cooks over-represented in web sources. Province attribution uncertainty. Sample size limitations for less-documented provinces. Temporal limitations — recipes collected in 2025 may not represent historical traditions. What the model cannot tell us about causation (it finds patterns, not explanations). Honest acknowledgment makes the paper more credible, not less.

8. Conclusion (500 words)

Restate findings, their significance, and implications. The OpenFlavorTH dataset contribution. Future research directions (historical evolution of Thai cuisine, cross-border comparisons with Lao/Malaysian cuisine, network dynamics over time).

References

Minimum 25 academic citations. Key required citations: Ahn et al. 2011 (flavor networks), West et al. 2013 (regional cuisines), Louvain community detection paper (Blondel et al. 2008), UMAP paper (McInnes et al. 2018), Moran's I (Moran 1950), at least 3 Thai food culture academic sources.

Data Availability Statement

All data available at HuggingFace: huggingface.co/datasets/flavormap/openflavorTH. All code available at github.com/flavormap. arXiv preprint at [arXiv ID].

Key citations she must read before writing

Citation	Why essential
Ahn, Y.Y. et al. (2011). Flavor network and the principles of food pairing. Scientific Reports, 1, 196.	The foundational paper for this entire field. Her paper builds directly on this methodology. Must cite and explain how her work differs (Thai-specific, geographic focus).
West, R. et al. (2013). From cookies to cooks: insights on dietary patterns via analysis of a recipe website. WWW 2013.	Studies regional cuisine variation computationally. Most directly comparable prior work. Her contribution is province-level granularity for Thai cuisine.
Blondel, V.D. et al. (2008). Fast unfolding of communities in large networks. Journal of Statistical Mechanics.	The Louvain community detection algorithm she uses. Must cite the original paper.

Citation	Why essential
McInnes, L. et al. (2018). UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. arXiv.	The UMAP algorithm. Cite the original arXiv paper.
Anselin, L. (1995). Local indicators of spatial association — LISA. Geographical Analysis, 27(2), 93-115.	Spatial autocorrelation methodology. Cite for Moran's I application.
Kasetsiri, C. (2003). Thai identity and its Southeast Asian implication. Asian Review.	Thai cultural geography — academic grounding for the cultural claims in the Discussion section.
Treesuwan, A. et al. (various years). Thai food composition database. Mahidol University Institute of Nutrition.	Thai nutritional and food science academic source. Establishes academic context for Thai ingredient research.

Writing process — practical guidance

- **Write the methods section first.** While the analysis is running, write down exactly what she did. The methods section is the easiest to write (it's just documenting what she did) and gets harder if delayed.
- **Results before discussion.** Write what the data shows before writing what she thinks it means. This prevents motivated reasoning.
- **One figure per finding.** Each major result (centrality table, community alignment heatmap, choropleth map, classifier confusion matrix) should have one corresponding figure in the paper.
- **All code in a GitHub repository with a README.** Papers that include a reproducibility link are reviewed more favorably at DH and computational venues.
- **arXiv first, conference second.** Post to arXiv in Month 11 regardless of whether a conference submission is ready. arXiv creates a timestamp that establishes the work's priority.

16. Dataset Release — OpenFlavorTH on HuggingFace

The OpenFlavorTH dataset is Pillar 3 (OpenScholarTH) applied to food. Releasing a documented public dataset serves three purposes: it is a permanent contribution that other researchers can build on; it signals she thinks beyond her own work; and the HuggingFace download count is a public, verifiable impact metric.

Dataset contents

- **Main recipe dataset:** 2,200+ recipes with all fields from Section 5's schema. CSV and JSON formats. Province-confidence field enables researchers to use only high-confidence data.
- **Canonical ingredient dictionary:** ~400 canonical ingredients with Thai names, English names, regional significance tags, and variant aliases. The dictionary itself is a standalone contribution.
- **Province metadata:** For each of the 77 provinces — region, geographic centroid, number of recipes, average distinctiveness score, top 5 most distinctive ingredients.
- **Network data:** The ingredient co-occurrence graph in GraphML and JSON formats. Node attributes include centrality metrics. Edge attributes include co-occurrence count and regional breakdown.
- **Analysis results:** UMAP embeddings, cluster labels, classifier predictions, Moran's I results — all reproducible from the published code.
- **Interview transcripts:** Anonymized and translated summaries (not full transcripts) from the fieldwork interviews, with province attribution.

HuggingFace data card requirements

A data card is a structured documentation file (YAML + Markdown) that HuggingFace requires for dataset releases. The card must describe the dataset's purpose, collection methodology, known biases, intended uses, and prohibited uses. Writing a thorough data card demonstrates data ethics awareness — a skill explicitly valued at iSchool and AI ethics-focused programs.

```
# Sample HuggingFace dataset card structure (README.md) --- language: th task_categories:
text-classification, graph-ml tags: - thai-cuisine - food-science - cultural-heritage -
network-analysis - geospatial pretty_name: OpenFlavorTH – Thai Regional Cuisine Dataset
size_categories: 1K-10K --- # OpenFlavorTH: Thai Regional Cuisine Dataset ## Dataset Description
2,247 Thai recipes from 77 provinces, collected via web scraping and fieldwork interviews (2025).
Includes province-level ingredient lists, cooking methods, and culinary network analysis data. ##
Collection Methodology [Describe scraping and interview methodology] ## Known Biases - Urban and
digitally active cooks over-represented in web data - Some provinces have fewer than 20 recipes
(marked province_confidence='low') - Historical recipes (pre-1970s) under-represented -
Muslim-influenced Southern cuisine may be under-represented ## Intended Uses Academic research on
Thai culinary geography, food science, cultural heritage preservation, computational humanities.
## Prohibited Uses Commercial food product development without separate licensing. Do not use to
make definitive claims about any specific cook or community.
```

17. 12-Month Implementation Timeline

This timeline assumes she starts FlavorMap in Month 1 while TunDee is in maintenance mode post-launch. The heavy build season is the summer break (Months 4–6). Analysis happens in Months 6–8. Writing in Months 9–11. Polish in Month 12.

Month	Focus	Key actions	Target output
1 (Now)	Setup + scraping pipeline	Set up project GitHub repo. Register flavormap.org on Porkbun. Begin scraping SorKorPor (best-structured source first). Build canonical ingredient dictionary starter (top 50 ingredients). Identify academic advisor (Chulalongkorn food science or cultural studies) through Dad.	Repo live. First 200 recipes scraped. Advisor meeting scheduled.
2	Scraping + first interviews	Complete SorKorPor scraping. Begin Wongnai and TAT scraping. Conduct first 5 fieldwork interviews (Bangkok-based diaspora cooks). Build canonical ingredient dictionary to 150 ingredients. Submit IRB ethics review request through Dad's Chulalongkorn contact.	600+ recipes. 5 interviews completed. Ethics review submitted.
3	Data collection sprint	Complete all major web scraping (Wongnai, Kapook, Thai Taste). Conduct 10–15 more interviews. Begin data cleaning pipeline — duplicate detection, ingredient extraction. Canonical ingredient dictionary complete (300+ ingredients). IRB approval received.	1,500+ recipes scraped. 15+ interviews. Cleaning pipeline running.
4–5 (Summer)	Deep build season	Complete remaining interviews (total 25–30). Complete full data cleaning pipeline. Province attribution for all recipes. Build complete PostgreSQL + PostGIS database. Build ingredient network (NetworkX). Run centrality analysis and community detection. Begin UMAP dimensionality reduction. Advisor meeting to review preliminary findings.	Dataset complete (2,200+ recipes). Network built. First results.
6	Geographic analysis	Geospatial analysis: download Thailand GADM shapefile, merge with province data. Run K-means and hierarchical clustering. Compute Moran's I and Mantel test. Build province-of-origin classifier. Analyze confusion matrix. Prototype choropleth map with Folium.	All analyses complete. Prototype map live.
7	Visualization build	Build production interactive visualization (Next.js + Mapbox + Sigma.js). Deploy to Vercel at flavormap.org. Stretch goal: begin food image classifier if ahead of schedule. Share preliminary findings with advisor for feedback.	flavormap.org live with all three views.
8	Paper writing begins	Write Methods section (easiest — document what was done). Write Results section (follow the five RQs). Create all paper figures (network visualization, choropleth, confusion matrix, UMAP plot). HuggingFace dataset card draft.	Methods + Results draft. All figures complete.
9	Paper completion	Write Introduction and Related Work (requires reading 15+ papers). Write Discussion and Conclusion. Full paper review with advisor. Release OpenFlavorTH dataset on HuggingFace.	Full paper draft. Dataset released on HuggingFace.
10	Submission	Address advisor feedback. Polish all figures to publication quality. Post arXiv preprint (cs.SI or cs.CY). Submit to primary venue (DH2026 or Journal of Cultural Analytics). Polish GitHub repository for public view.	arXiv preprint live. Paper submitted.
11–12	Polish + admissions	Update visualization with any improvements from advisor feedback. Write FlavorMap section of Common App activities. Compile all materials for university application. If paper accepted: update application materials. If not yet accepted: 'submitted to [venue], under review' is sufficient.	All application materials ready. arXiv preprint cited.

Engineering effort estimate

Component	Hours
Data collection (scraping setup + running + validation)	60 hrs
Fieldwork interviews (preparation, conducting, transcription, coding)	50 hrs
Data cleaning and standardization pipeline	70 hrs
Network analysis (NetworkX, centrality, community detection)	40 hrs
Geographic analysis (PostGIS, clustering, Moran's I, Mantel)	40 hrs
Province classifier (feature engineering, model training, evaluation)	30 hrs
Interactive visualization (Mapbox + Sigma.js + Next.js frontend)	60 hrs
Research paper writing (with advisor)	50 hrs
HuggingFace dataset release and documentation	15 hrs
Food image classifier (stretch goal only)	50 hrs (additional)
TOTAL (without stretch goal)	~415 hrs

18. Revenue and Social Enterprise Connection

FlavorMap generates revenue for KrueChai Foundation through data licensing — not through user subscriptions. The dataset she builds has commercial value to organizations that benefit from understanding Thai culinary geography. This is the most honest and sustainable revenue model for a research project.

Three revenue streams from FlavorMap data

Stream	Who pays	What they get	Revenue estimate
Food industry data licensing	Thai food brands (MaMa, Singha, Oishi, Thai Union), restaurant chains (Coca, MK), food delivery (GrabFood, Foodpanda)	Province-level ingredient trend data for product development, regional menu planning, and marketing localization. Delivered as quarterly reports.	200,000–500,000 THB/year from 1–2 commercial partners
Tourism sector licensing	Tourism Authority of Thailand (TAT), Airbnb Experiences (food tourism), hotel chains, travel magazines	'Culinary tourism map' — which province to visit for which ingredient tradition. Interactive version of flavormap.th licensed for tourism marketing materials.	100,000–300,000 THB from TAT partnership or equivalent
Cultural heritage grants	Office of National Culture Commission, UNESCO Bangkok, SCB Foundation (funds Thai cultural preservation)	The recipe documentation work qualifies as intangible cultural heritage preservation. Apply for project grants rather than ongoing licensing.	200,000–600,000 THB (grant, one-time)
TOTAL			500,000–1,400,000 THB

The GSEA social enterprise pitch: FlavorMap is the data infrastructure arm of KrueChai Foundation's TunDee ecosystem. TunDee provides scholarship access. FlavorMap documents the culinary heritage of the communities TunDee serves. Revenue from FlavorMap data licensing funds TunDee's free tier for rural students. Both projects serve the same population; one generates revenue that subsidizes the other.

19. How FlavorMap Connects to TunDee

The most important admissions question about a two-project portfolio is not 'what did you build?' but 'why do these two things belong together?' The answer must be intellectual, not just thematic.

The shared thesis

<i>"Culture is infrastructure — the data we trust, the memories we preserve, and the systems we build are all forms of the same thing. I build infrastructure that redistributes opportunity to those whom existing systems leave out."</i>		
Dimension	TunDee	FlavorMap
What is invisible	3,000 scholarships invisible to rural students	Provincial culinary traditions invisible in aggregate 'Thai food' narrative
Who is excluded	Rural students systematically excluded from scholarship information	Provincial communities erased from their own culinary heritage by centralization
What the algorithm finds	Which student matches which scholarship — against the bias of historical data	Which province encodes which culinary tradition — against the homogenizing narrative of 'Thai food'
What gets redistributed	Access to scholarship opportunity	Visibility of cultural identity
The data science method	Fairness-aware recommendation + causal inference	Network analysis + geographic clustering + classifier
The human story	Ploy in Surin who didn't know the scholarship existed	The elderly cook in Surin whose pla ra recipe was never written down
The public output	tundee.org — 8,000+ users finding scholarships	flavormap.org — interactive map of Thai culinary geography

The Surin connection — the essay's unifying moment

Surin province appears in both projects. TunDee's prototype user (Ploy) is from Surin. Isaan cuisine (which Surin exemplifies) is one of FlavorMap's most analytically interesting regions — highest distinctiveness scores, strongest pla ra signal, greatest distance from Bangkok culinary norms. The essay can say: *'I built TunDee for Ploy in Surin, who didn't know the scholarship existed. Then I built FlavorMap and discovered that Surin's cuisine — the food her grandmother cooks — is one of the most distinctive in Thailand. Both systems are doing the same thing: making visible what the dominant systems made invisible.'* This is the unifying moment.

20. The Admissions Story — What to Say and How

Common App activity description (150 characters)

Position: Founder & Lead Researcher

Organization: FlavorMap (แผนที่รสชาติ) · KrueChai Foundation

Description (148 chars): Built first computational map of Thai regional cuisine: ingredient network analysis across 2,200+ recipes and 77 provinces. Interactive visualization at flavormap.org.

Essay paragraph (the culture pillar)

The FlavorMap paragraph replaces the KhruNat paragraph in the essay. Same intellectual position — 'culture is infrastructure, and I map what the dominant systems make invisible' — but grounded in cooking, which is authentically hers.

"The second infrastructure I built was for memory. My grandmother cooks a tom yum that tastes nothing like Bangkok restaurant tom yum — sharper, with pla ra instead of fish sauce, with wild herbs that have no English name. For years I assumed this was nostalgia. Then I built FlavorMap and analyzed 2,200 Thai recipes across all 77 provinces. The ingredient network showed her version as a statistically distinct cluster — not personal taste, but centuries of Isaan trade routes encoded in a soup. Thailand describes itself as one cuisine. The data shows 77. FlavorMap makes the invisible ones visible."

Interview talking points

- **When asked 'what did you find?':** 'The data confirmed that Southern Thai cuisine is most distinct — it uses fresh turmeric, coconut milk, and a completely different set of chilies than anywhere else. But the most surprising finding was Northeastern Thailand — the Isaan provinces showed the highest pla ra signal, which encodes the ancient Khmer trade routes that predated Bangkok by centuries.'
- **When asked 'what was the hardest part?':** 'Province attribution. Most online recipes don't say which province they're from — they just say "Isaan style" or "Northern curry." I built a four-tier attribution system and was honest about confidence levels. The paper only uses high and medium confidence attributions for the main analysis.'
- **When asked 'what was surprising?':** 'The classifier predicted the correct region 78% of the time, which is better than I expected. But it was consistently confused between Surin and Sisaket [two adjacent Isaan provinces]. That confusion is itself a finding — it suggests they share a culinary tradition that cuts across provincial administrative lines.'
- **When asked 'how does this connect to TunDee?':** 'Both projects are about making invisible things visible. TunDee maps scholarships that rural students can't find. FlavorMap maps culinary traditions that get erased when we describe everything as "Thai food." Same methodology, same thesis, different domain.'

21. Risks and Mitigations

Risk	Severity	Mitigation
Province attribution too uncertain — most recipes end up province=null	HIGH	Use the four-tier attribution strategy. Run analysis at regional level (4 classes) when province-level data is insufficient. Regional analysis is publishable and still answers the core research question.
Ingredient canonicalization creates false equivalences — over-aggregating distinct ingredients	HIGH	Have the academic advisor (food science background) review the canonical ingredient dictionary before analysis begins. Maintain the NOT_same_as field in the dictionary to flag common conflations.
Network analysis finds no significant regional structure — null result	MEDIUM	A null result (Thai cuisine is more homogenized than expected) is publishable and intellectually honest. The paper becomes about the limits of geographic structure in Thai food, which is also interesting. She should not falsify or discard null results.
Too few recipes per province for reliable analysis (<20 per province)	MEDIUM	Focus interview fieldwork on provinces with lowest web-scraped coverage. Aggregate under-represented provinces into regional groups for the classifier. Document which provinces have insufficient data in the paper's limitations section.
Web scraping blocked by target sites	MEDIUM	Check robots.txt first. Use respectful rate limiting (1 request/second). If blocked, use manual data entry for that source or find an alternative. SorKorPor (government site) and TAT are most scrape-friendly.
Academic advisor hard to find or not engaged enough	MEDIUM	Approach Chulalongkorn's Faculty of Liberal Arts or Faculty of Arts (Thai studies, cultural geography) and Faculty of Science (food science). Dad introduces. Offer co-authorship on the paper — this is the right incentive for an academic.
Interactive visualization too complex to build well	LOW-MED	Start with Folium prototype (pure Python, 2 hours to build basic map). Ship the Folium version first. Upgrade to Mapbox + Sigma.js only if prototype is working. A simple working visualization beats a complex broken one.
Time conflict with TunDee maintenance and school	MEDIUM	Heavy FlavorMap work happens in summer (months 4–6) when TunDee is in stable maintenance. During school terms: max 5 hrs/week on FlavorMap. Academic GPA is always the first priority.
Paper rejected from all venues	LOW	arXiv preprint + HuggingFace dataset release are sufficient for admissions. 'Submitted to Digital Humanities 2026, under review' is legitimate. Even a workshop paper is strong for a high school first author.

22. First 30 Days — Concrete Actions

The project does not exist until these actions are done. Do them in order. The goal of the first 30 days is not to have built anything impressive — it is to have built the foundation that makes everything else possible.

Day	Action	Owner	Done when
1–2	Register flavormap.org on Porkbun (\$6.88). Set up GitHub repo (github.com/flavormap). Create a README with the project description and research questions. This makes the project real and timestamped.	Daughter	Domain registered. Repo public with README.
3–5	Read the three foundational papers before writing a single line of code: Ahn et al. 2011 (flavor networks), West et al. 2013 (regional cuisine analysis), Blondel et al. 2008 (Louvain community detection). Take notes on their methodology. These papers shape every design decision.	Daughter (solo reading)	Notes summarizing each paper's methodology written in research notebook.
6–7	Request academic advisor introduction through Dad. Target: Chulalongkorn Faculty of Liberal Arts, Faculty of Arts, or Faculty of Science (food science). Email should explain the project, the research questions, the paper target venue (DH2026), and offer co-authorship.	Daughter drafts; Dad introduces	Email sent to at least 2 potential advisors.
8–10	Build the first scraper for SorKorPor (กรมส่งเสริมการเกษตร). This is the highest-quality, most province-specific Thai recipe source. Target: 300 recipes scraped and stored in PostgreSQL by Day 10. Use BeautifulSoup. With coding mentor.	Daughter + coding mentor	300 recipes in database. Scraper code committed to GitHub.
11–14	Build canonical ingredient dictionary starter (top 100 ingredients). Create a Google Sheet with columns: canonical_id, canonical_name_th, canonical_name_en, category, regional_significance, aliases. Fill in manually for the 100 most common Thai ingredients. ISB Club members can help.	Daughter + 2 ISB Club members	100-ingredient dictionary in spreadsheet, exported to JSON.
15–17	Conduct first 3 fieldwork interviews (Bangkok-based). Target: diaspora home cooks from Chiang Mai, Isaan, and Southern Thailand, sourced through ISB staff network or community associations. Mom present. Record with consent. Note the 5 dishes each cook considers most distinctively from their province.	Daughter + Mom	3 interviews completed. Transcription notes in research notebook.
18–21	Build ingredient extraction NLP pipeline. Takes raw Thai recipe text (from scraped HTML) and extracts a clean list of canonical ingredient names. Use Claude API with a structured prompt. Test on 50 scraped recipes from SorKorPor. Manually validate 100% of outputs.	Daughter + coding mentor	Pipeline running. 300 recipes fully cleaned with canonical ingredient lists.
22–25	Submit IRB ethics review request through Dad's Chulalongkorn contact. One-page form describing the interview methodology, consent procedures, anonymization plan. Essential for paper to be publishable at academic venues.	Dad facilitates; daughter completes form	IRB request submitted.
26–30	Build first NetworkX ingredient network from the 300 SorKorPor recipes. Even with 300 recipes, a preliminary network is visible. Compute basic centrality metrics. Share a visualization screenshot with the academic advisor at the first meeting. This shows momentum and gets early feedback on research direction.	Daughter + coding mentor	First network visualization produced. Advisor meeting held.

Final note: FlavorMap is not a product. It is a research project with a public face. The distinction matters. Products succeed when users adopt them. Research projects succeed when they answer a genuine question honestly. Her success metric is not 'how many people visit flavormap.org' — it is 'does the data show something nobody knew before?' If she approaches this as research, the paper will be rigorous and the visualization will be honest. If she approaches it as a product, both will be shallow. She loves cooking. She is curious about where food comes from. Let that curiosity drive every decision in this project.